

STYLEMATE - PERSONALIZED AI FASHION ASSISTANT

(SMART DIGITAL WARDROBE)

25-26J-302

Project Proposal Report

Abirajh A - IT22066848

B.Sc. (Hons) in Information Technology Specializing in Information
Technology

Department of Information Technology Faculty of computing

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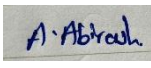
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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher Learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text

Name	Student ID	Signature
Abirajh A	IT22066848	

The above candidate is carrying out research for the undergraduate dissertation under my supervision

.....

Signature of the supervisor

(Ms. Anjalie Gamage)

Date.....

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Signature of the Co-supervisor

(Prof. Samantha Rajapaksha)

Date.....

Abstract

The Smart Digital Wardrobe addresses the common problem of wardrobe redundancy and inefficiency, where users frequently repeat outfits or purchase unnecessary clothing. The purpose of this study is to develop an AI-powered system that enables users to make better use of their existing wardrobe while providing personalized, practical, and context-aware outfit recommendations. The system identifies clothing type and colour from user-uploaded images, with the fabric type provided manually by the user. It then integrates user profiles, such as occupation (student or office worker), and real-time weather data to suggest daily outfits that are both stylish and suitable for environmental conditions.

The basic design of the study involves image processing and machine learning for clothing recognition, combined with a recommendation engine that generates mix-and-match outfits, group or couple coordinated suggestions, and travel wardrobe plans based entirely on the clothes the user already owns. The system ensures minimal repetition, optimizes outfit combinations, and supports efficient packing for trips by suggesting the number and type of outfits required for each day.

While this proposal focuses on system development and design, the expected outcomes include improved user satisfaction, reduced clothing redundancy, and enhanced confidence and style awareness through context-aware recommendations. The system demonstrates how AI, image recognition, and personalized recommendation algorithms can be applied to a practical, everyday problem in wardrobe management. The results of this study are anticipated to show that users can achieve optimized wardrobe utilization and better planning for events and trips, while also promoting sustainable fashion practices by reducing unnecessary clothing purchases.

Keywords: Smart Digital Wardrobe, AI-based fashion recommendation, clothing recognition, personalized outfit suggestions, mix-and-match, wardrobe optimization.

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List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
ML	Machine Learning
API	Application Programming interface
UI	User interface
UX	User Experience

1.0 INTRODUCTION

1.1 Background and Literature

The fashion and apparel industry has increasingly turned to AI-powered recommendation systems to help users make better clothing choices. Previous research and commercial solutions have explored personalized outfit suggestions, leveraging machine learning and image processing techniques to analyse user preferences, clothing attributes, and contextual information such as weather or occasion [1] [3]. Systems like Vue.ai and Dataiku's AI Outfit Recommendation System demonstrate the capability of AI to provide personalized clothing suggestions by analysing user profiles and wardrobe data [1], [2].

To implement a system like the Smart Digital Wardrobe, it is essential to master image recognition, clothing attribute extraction (type, colour, fabric), and recommendation algorithms such as collaborative filtering and content-based filtering. Understanding context-aware recommendation techniques including weather integration, occupation-based personalization, and group/couple coordination is also critical [4], [5].

The state of the art today includes AI-powered solutions capable of generating outfit suggestions based on user preferences, past behaviour, and visual search of products. However, most existing systems are limited in that they either focus on recommending new items instead of optimizing the user's existing wardrobe or lack group and travel wardrobe planning functionalities [1], [2], [3].

Previous attempts to solve wardrobe-related problems primarily rely on either product-centric recommendations or trend-based suggestions, without fully integrating user wardrobe analysis, weather context, and mix-and-match personalization [4], [5]. The Smart Digital Wardrobe builds upon these works by combining image-based clothing recognition, user profile integration, weather-aware recommendations, and travel wardrobe planning, providing a holistic and practical solution for modern wardrobe management.

1.2 Research Gap

Despite the growing adoption of AI in fashion recommendation systems, current solutions largely focus on product-centric suggestions or trend-based recommendations, often ignoring the user's existing wardrobe. Systems like Vue.ai and Dataiku's AI outfit recommendation platform provide personalized clothing suggestions, but they primarily recommend new purchases rather than optimizing the utilization of clothes that users already own [1], [2]. Similarly, existing AI systems consider basic user preferences but fail to fully integrate contextual factors such as weather conditions, user occupation, or planned trips, limiting the practical usability of recommendations [3], [4].

Additionally, most commercial systems lack group or couple coordination features, which are crucial for events, social gatherings, or family trips. They also do not provide mix-and-match recommendations to avoid repeated outfit usage, which is a significant issue for users seeking wardrobe efficiency and sustainable fashion practices [4], [5]. Current research in fashion recommendation systems often emphasizes algorithmic accuracy or product matching but does not comprehensively address the holistic integration of user wardrobe, context, and personalization.

The Smart Digital Wardrobe addresses this gap by combining clothing type and colour recognition, user-provided fabric data, user profiles, and real-time weather information to generate daily outfit suggestions, travel wardrobe plans, and group/couple coordinated recommendations. This approach enhances practicality, personalization, and sustainability, offering a solution that goes beyond what existing systems provide, bridging the gap between AI-based fashion recommendations and real-world wardrobe management needs.

1.3 Research Problem

Managing personal wardrobes efficiently is a common challenge for many users, leading to repetitive outfit choices and unnecessary clothing purchases. People often struggle to plan daily outfits based on their existing clothing, especially when considering weather conditions, occupation, or planned trips. Current fashion recommendation systems focus primarily on recommending new clothing items or following general fashion trends. While these systems offer some level of personalization, they fail to incorporate the user's existing wardrobe and rarely consider contextual factors like weather, occupation, or events, which are essential for practical, real-world outfit planning.

Another limitation of existing systems is the lack of mix-and-match suggestions and group or couple coordination, which are important for social events, trips, or family occasions. Users often must manually plan outfits for such situations, resulting in inefficient wardrobe utilization and reduced confidence in styling choices.

The research problem addressed in this study is the development of an AI-powered Smart Digital Wardrobe capable of analysing user-uploaded clothing images to identify clothing type and colour, while incorporating user-provided fabric information, profile data, and weather forecasts. The system aims to generate personalized outfit suggestions, daily mix-and-match combinations, travel wardrobe plans, and coordinated outfits for groups or couples. This approach not only maximizes the use of existing wardrobe items but also provides a context-aware, practical, and sustainable fashion recommendation system, bridging the gap between traditional AI outfit recommendations and real-world wardrobe management needs.

2.0 OBJECTIVE

2.1 Main Objective

To develop an AI-powered Smart Digital Wardrobe that provides personalized, context-aware outfit recommendations using the user's existing wardrobe, improving wardrobe utilization, reducing outfit repetition, and enhancing user confidence in daily dressing.

2.2 Specific Objective

The following are the sub-objectives of conducting this research.

- Detect clothing type and colour from user-uploaded images.
- incorporate user-provided fabric information and profile details (e.g., student or office worker) into outfit recommendations.
- Provide weather-appropriate outfit suggestions for daily wear.
- Generate mix-and-match outfit combinations to prevent repetition and optimize wardrobe usage.
- Recommend coordinated outfits for groups or couples.
- Plan travel wardrobes, suggesting the number of outfits and daily combinations based on location, weather, and trip duration.

3.0 METHODOLOGY

3.1 Requirements Gathering

Requirement gathering was through performing an extensive analysis of past research conducted throughout recent years, identification and analysis of the existing systems, as well as reading through a variety of online resources. Also, some real-world scenarios were used to figure out this research problem and requirement gathering.

3.1.1 Past Research Analysis

Previous studies and commercial systems have explored AI-based fashion recommendations, but most have limitations that the Smart Digital Wardrobe addresses. Platforms like Dataiku's AI Outfit Recommendation demonstrate the potential of machine learning to suggest outfits based on clothing attributes and user preferences. However, these systems primarily focus on recommending new purchases, neglecting the user's existing wardrobe. Similarly, Vue.ai provides personalized outfit suggestions by analysing user behaviour and profiles, but it lacks contextual awareness, such as weather conditions, travel plans, or group/couple coordination.

Research in fashion recommendation techniques emphasizes methods like collaborative filtering, content-based filtering, and deep learning for image recognition. While effective for matching preferences and identifying clothing attributes, these methods are seldom integrated into a comprehensive wardrobe management system. Existing systems rarely offer mix-and-match suggestions, travel wardrobe planning, or coordinated outfits for groups and couples. Additionally, weather-aware and context-sensitive recommendations are often missing, reducing the practical usability of these solutions.

The Smart Digital Wardrobe builds upon these insights by combining clothing type and colour recognition, fabric input, user profile integration, weather-based suggestions, mix-and-match combinations, and travel wardrobe planning. This approach addresses gaps in previous research, providing a practical, personalized, and context-aware wardrobe management system that optimizes existing clothing usage while enhancing style, convenience, and sustainability.

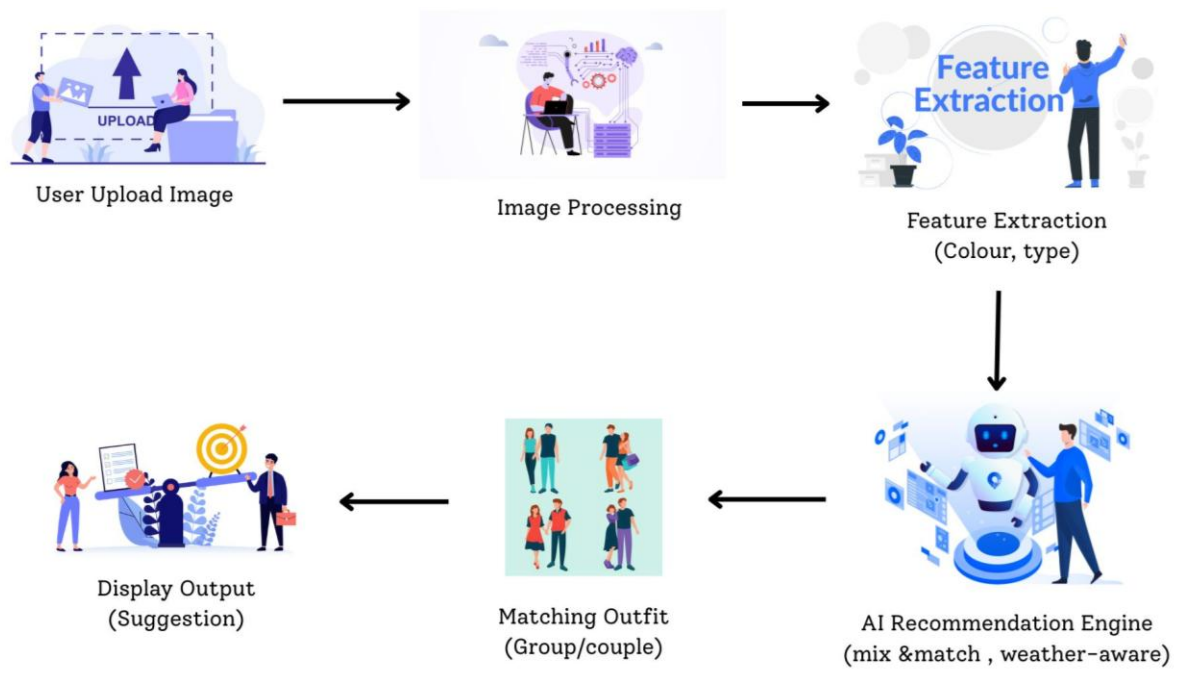


Figure 1 : System Diagram

3.2 Feasibility Study

3.2.1 Technical Feasibility Study

3.2.1.1 Knowledge on Technology

To develop the proposed solution, the knowledge in below mentioned technologies should be wanted.

1. Machine Learning (ML):
 - Purpose: Enables the system to recognize clothing types and colours from user-uploaded images.
 - Knowledge Required: Understanding Convolutional Neural Networks (CNNs) for image classification, feature extraction, and model training from scratch.
2. Image Processing & Computer Vision:
 - Purpose: Prepares and analyses images to accurately detect clothing attributes.
 - Knowledge Required: Familiarity with libraries like OpenCV or PIL, image preprocessing techniques (resizing, normalization, colour extraction), and feature detection.
3. Recommendation Systems:
 - Purpose: Suggests mix-and-match outfit combinations, travel wardrobe plans, and coordinated group/couple outfits.
 - Knowledge Required: Understanding collaborative filtering, content-based filtering, and context-aware recommendation techniques.
4. Weather Integration:
 - Purpose: Provides weather-appropriate outfit suggestions.
 - Knowledge Required: Familiarity with API integration (e.g., OpenWeather) to fetch real-time weather data and use it to adjust clothing recommendations.
5. Full-Stack Web Development:
 - Purpose: Implements the user interface and backend system for seamless interaction.
 - Knowledge Required: Proficiency in frontend development (React.js), backend development (Flask/FastAPI), and database management for storing user profiles and wardrobe data.
6. UI/UX Design:
 - Purpose: Ensures the system is user-friendly and intuitive.
 - Knowledge Required: Principles of responsive design, Material-UI, and prototype development (Figma) for a smooth user experience.

3.2.1.2 Knowledge on Tools

In order to develop the proposed optimization model, all members should have quite an in-depth understanding of the development tools and project management, other supportive tools.

3.2.1.3 Data Collection Knowledge

Data Collection:

- Collect clothing images from online sources, fashion websites, and user uploads.
- Gather user inputs such as fabric type, occupation (student/office), and profile information.
- Label images with clothing attributes: clothing type, colour, and fabric.

Data Preprocessing:

- Data Cleaning: Remove duplicates, low-quality images, and incorrect labels.
- Data Transformation: Resize images, normalize pixel values, encode categorical attributes.
- Data Reduction: Select representative images or crop images to focus on clothing items.
- Data Integration: Combine multiple datasets into a structured dataset ready for training.

3.2.2 Schedule Feasibility

This component represents an independent solution and will be developed and implemented in accordance with the established project timeline.

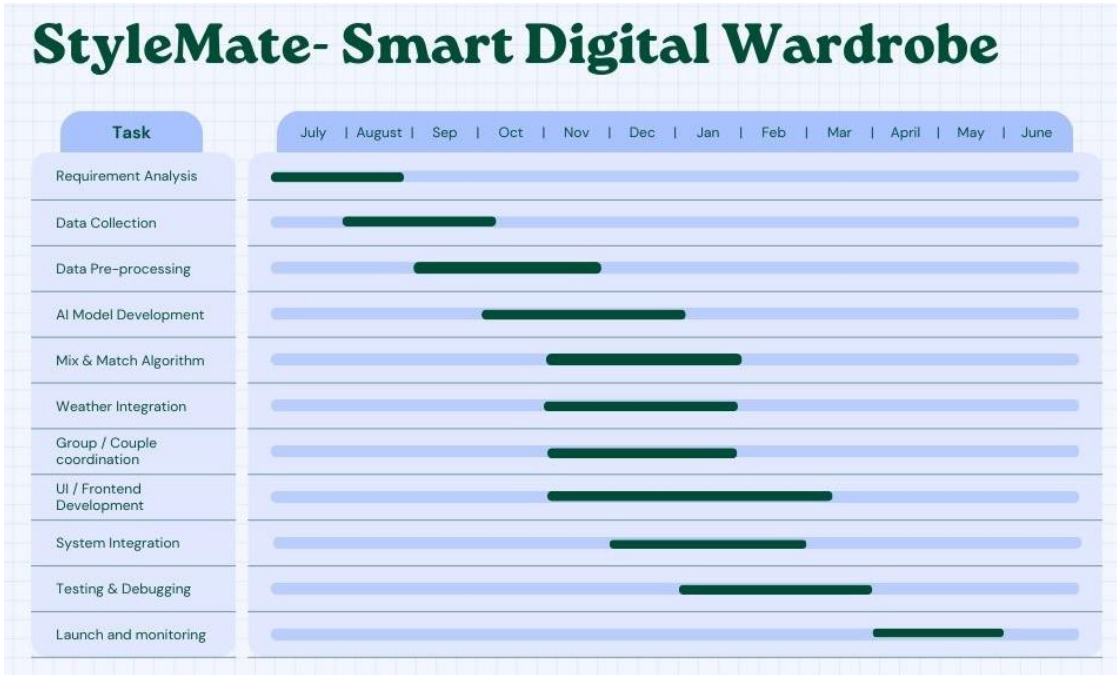


Figure 2 : Gannt chart

3.3 System Analysis

3.3.1 Software solution Approach

1. User Input Module:

- Allows users to upload images of clothing items.
- Collects additional information such as fabric type, occupation (student/office worker), and trip details.

2. Data Processing Module:

- Performs image preprocessing (resizing, normalization, and cropping) to prepare images for analysis.
- Stores labelled clothing attributes (type, colour, fabric) in a structured database.

3. Clothing Recognition Module:

- Utilizes Convolutional Neural Networks (CNNs) to detect clothing type and color.
- Ensures accurate recognition for mix-and-match outfit recommendations.

4. Recommendation Engine:

- Generates personalized daily outfit suggestions based on user profile and wardrobe.
- Integrates weather data for weather-appropriate recommendations.
- Supports group or couple outfit coordination and travel wardrobe planning.

5. User Interface Module:

- Provides a responsive web interface for seamless interaction.
- Displays recommended outfits, mix-and-match suggestions, and travel wardrobe plans clearly and intuitively.

This approach ensures that the system is user-centric, scalable, and practical, providing a comprehensive solution for modern wardrobe management while promoting sustainable clothing usage.

3.3.2 Tools and Technology

Programming Languages: Python, JavaScript

Frameworks & Libraries:

- Machine Learning / AI: TensorFlow, Keras, OpenCV
- Web Development: Node.js, Express.js, Flask, FastAPI, React.js
- UI Components: Material-UI (MUI)

Database Systems: MongoDB

UI/UX Design Tools: Figma

Diagramming Tools: Draw.io

Integrated Development Environments (IDE): VS Code

Version Control Systems (VCS): Git (GitHub, GitLab)

Collaboration Tools: Microsoft Teams, WhatsApp

Testing Tools:

- JavaScript Unit Testing: Jest
- Python Unit Testing: unittest, pytest
- API Testing: Postman

Containerization & Orchestration: Docker, Kubernetes

Automation & Configuration Tools: Jenkins, Ansible, Chef

Code Quality Assurance: SonarQube

Project Management Tools: Microsoft Planner

3.4 Project Requirements

3.4.1 Functional Requirements

The functional requirements for the proposed model are as follows:

- Upload clothing images and input fabric type and profile details.
- Automatically detect clothing type and color.
- Provide daily outfit suggestions with mix-and-match combinations.
- Integrate weather data for suitable outfit recommendations.
- Suggest coordinated outfits for groups or couples.
- Plan travel wardrobes using existing clothes.

3.4.2 Non-Functional Requirements

Key non-functional requirements considered in developing the proposed model are:

- Usability
- Reliability
- Availability
- Scalability
- Accuracy
- Privacy
- Performance

3.5 Testing

When it comes to testing the research outcome, it plays a crucial role in the success of this project. To test the functionality of the proposed Smart Digital Wardrobe, the initial plan is to test the system among team members and friends. This will act as unit testing and system testing to ensure that clothing recognition, mix-and-match suggestions, weather integration, and travel wardrobe planning work correctly.

In the final phase of the research, the system will be tested with a selected group of students, office workers, and fashion enthusiasts in a local environment and deployed web application. This will act as acceptance testing, including Alpha and Beta testing, to validate the usability, accuracy, and practical functionality of outfit recommendations.

Most of the testing will be manual, where users interact with the system and provide feedback on outfit suggestions. Additionally, internal functions and assert statements will be used to verify the correctness of clothing detection, colour recognition, recommendation logic, and integration with weather data.

This multi-level testing approach ensures that the Smart Digital Wardrobe is robust, user-friendly, and reliable before full-scale deployment.

4.0 COMMERCIALIZATION

This is a Smart Digital Wardrobe system, and the target audiences of this application are students, office workers, and fashion-conscious individuals. Therefore, this solution will be needed by anyone looking to optimize wardrobe usage, avoid repeated outfits, and plan personalized outfits.

Currently, this system focuses on managing existing wardrobes, weather-aware outfit suggestions, mix-and-match combinations, group/couple coordination, and travel wardrobe planning. The proposed solution will be integrated as a web and mobile application, and these features are unique and not widely available in existing fashion recommendation systems. Therefore, we can provide these components according to user needs through a subscription-based methodology. By comparing with competitors, these features are innovative and practical, offering significant added value.

We could also integrate the travel wardrobe planning and group/couple coordination features for premium users of the application. For this, partnerships with fashion retailers, e-commerce platforms, or lifestyle apps could be explored. Alternatively, these features can be offered on a subscription basis with authorized access, providing both B2C and B2B opportunities for commercialization.

5.0 BUDGET

Category	Description	Estimated Cost (LKR)
Hosting & Cloud	Basic cloud server / GPU rental for 6 months	30,000 .00
Domain & Web Hosting	Domain (stylemate.lk) + basic web hosting	10,000.00
Publication	Conference/journal fees + printing/binding	12,500 .00
Deployment	Cloud deployment of prototype (3–4 months)	15,000.00
Marketing (Optional)	Posters, brochures, small scale promotions	20,000.00
Total Estimated		87,500.00

Table 1:cost management

This amount may differ according to the economic crisis.

6.0 SUMMARY

The Smart Digital Wardrobe is an AI-powered, context-aware wardrobe management system designed to optimize the use of existing clothing. By analysing clothing type and colour from user-uploaded images and integrating fabric type, user profiles, and weather data, the system provides personalized daily outfit recommendations, mix-and-match combinations, and coordinated outfits for groups or couples. Additionally, it assists in travel wardrobe planning, suggesting the number of outfits and daily combinations based on trip location and weather.

This system addresses gaps in existing fashion recommendation platforms, which often focus on recommending new items rather than leveraging the user's wardrobe. By combining machine learning, image processing, recommendation algorithms, and user-centric design, the Smart Digital Wardrobe offers a practical, sustainable, and user-friendly solution. It enhances user convenience, confidence, and style awareness, while promoting efficient wardrobe utilization.

In conclusion, the Smart Digital Wardrobe demonstrates the potential of AI and context-aware systems to transform everyday fashion management, offering a holistic, personalized, and scalable solution for modern users.

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